

ST CROIX LIME TREE BAY, Virgin Islands, U.S.

WMO: 997349

Lat: 17.695N

Lon: 64.754W

Elev: 10

StdP: 101.20

Time Zone: -4.00 (W04)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	23.3	23.8	N/A	N/A	N/A	N/A	N/A	N/A	8.2	25.9	7.5	26.1	3.3	80	0.411

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.2	31.2	N/A	30.9	N/A	30.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3.5	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	7.0	6.3	DB	21.9	31.8	0.5	0.4	21.6	32.1	21.3	32.3	21.1	32.6	20.8	32.9	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	26.0	25.8	26.0	26.7	27.5	28.4	28.6	28.9	28.8	28.6	27.7	26.7
	DBStd	1.37	0.63	0.58	0.87	0.76	0.91	0.67	0.64	0.81	0.73	0.87	0.84	0.72
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6382	496	444	496	501	543	553	577	586	564	577	530	517
	CDD18.3	3341	238	211	237	251	284	303	318	328	314	318	280	258
	CDH23.3	36256	1935	1661	1937	2390	3121	3666	3932	4155	3974	3957	3097	2431
CDH26.7	10270	147	96	173	344	772	1287	1471	1688	1591	1513	831	358	
Wind	WSAvg	3.5	3.5	3.5	3.5	3.6	3.5	3.8	4.0	3.8	3.0	3.2	3.1	3.4
Precipitation	PrecAvg	1016	55	40	39	67	103	59	70	97	126	121	144	106
	PrecMax	2064	191	83	134	158	431	187	211	321	476	400	510	295
	PrecMin	544	0	4	0	11	9	6	21	20	32	8	38	10
	PrecStd	325	43	26	30	41	114	53	44	76	94	84	122	70
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.6	28.3	29.2	29.7	30.6	30.9	31.2	32.3	31.6	31.5	30.5	29.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	28.1	27.9	28.5	29.1	29.9	30.5	30.7	31.4	31.1	31.0	30.1	29.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	27.8	27.6	28.0	28.6	29.5	30.2	30.4	30.8	30.7	30.6	29.7	28.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	27.5	27.3	27.6	28.2	29.1	29.9	30.1	30.5	30.4	30.3	29.4	28.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	3.2	3.1	3.1	3.0	2.9	2.9	3.1	3.2	3.3	3.3	3.5	3.3
		MCDBR	3.3	3.3	3.2	3.1	3.0	2.9	3.3	3.3	3.4	3.5	3.5	3.4
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		Clear-Sky Solar Irradiance	taub	0.370	0.375	0.380	0.405	0.449	0.514	0.518	0.478	0.437	0.392	0.378
taud	2.583		2.563	2.545	2.496	2.386	2.163	2.147	2.299	2.440	2.587	2.620	2.607	
Ebn at Noon	918		929	931	902	852	790	789	827	864	901	904	903	
Edh at Noon	94		101	105	111	123	152	155	134	115	97	90	90	
All-Sky Solar Radiation	RadAvg	4.95	5.75	6.34	6.64	6.41	6.44	6.55	6.46	6.03	5.34	4.73	4.58	
	RadStd	0.12	0.10	0.22	0.34	0.39	0.36	0.29	0.29	0.37	0.36	0.30	0.20	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	+0.23	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	+79	+80

Nomenclature: See separate page